PUSHFORWARD ALONG A THICKENING IDENTIFIES RELATIVE CYCLES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Checking the relative cycle condition after a closed immersion (relative-cycles chapter, Lemma lemma-check-after-closed, canonical tag 0H54). cited at proof lines 1474-1476. Gives alpha in z(X/S, r) if and only if i_* alpha in z(X'/S, r).

Source reference. Original lemma 0H54: sources/stacks-chow/tags.txt, source lines 21374–21374. Cited at proof lines 1474-1476: equivalence of the relative cycle condition under i_*.

Lemma 0.2. *Let S be a locally Noetherian scheme. Let $i : X \to X'$ be a thickening of schemes locally of finite type over S. Let $r \geq 0$. Then $i_* : z(X/S, r) \to z(X'/S, r)$ is a bijection.*

Proof. Since $i_s : X_s \to X'_s$ is a thickening it is clear that i_* induces a bijection between families of r-cycles on the fibres of X/S and families of r-cycles on the fibres of X'/S. Also, given a family α of r-cycles on the fibres of X/S $\alpha \in z(X/S, r) \Leftrightarrow i_*\alpha \in z(X'/S, r)$ by Lemma 0H54. The lemma follows. □

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